

1 Interprocess communication

1.1 Producer-consumer buffer

Theorem 1.1. *If access to the buffer is protected by mutual exclusion, the number of elements remains between zero and the capacity N .*

$$0 \leq n_{\text{buffer}} \leq N \tag{1.1}$$



Figure 1.1. Data flow through the shared buffer

Table 1.1. Buffer semaphore state	
Semaphore	Initial value
mutex	1
empty	N
full	0

```
1 def can_consume(items):  
2     return items > 0
```

Listing 1.1. Checking buffer availability

Input: A non-empty shared buffer
Output: The oldest element

```
1 wait(full)  
2 wait(mutex)  
3  $x \leftarrow$  remove element  
4 signal(mutex)  
5 signal(empty)  
6 return  $x$ 
```

Algorithm 1.1. Removing an element from the buffer

The invariant in theorem 1.1, expressed by equation (1.1), prevents overflow and underflow. Figure 1.1 summarizes the flow, while table 1.1, listing 1.1, and algorithm 1.1 describe initialization and removal.